

MUHAMMAD AKBAR

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EDUCATION

NED University of Engineering and Technology

Bachelor of Science (B.S.) in Computer Science and Information Technology

Karachi, Pakistan

Oct 2022 – Jun 2026 (Expected)

Bahria College Karsaz

Pre-Engineering

Karachi, Pakistan

Aug 2020 – July 2022

EXPERIENCE

Integration Xperts

Data Science Intern

Karachi, Pakistan

June 2025 – August 2025

- Assisted in developing computer vision solutions, including factory compliance monitoring and a facial recognition-based access system, demonstrating adaptability to diverse projects.
- Implemented automated testing using Pytest to evaluate age and gender classification models on CCTV image datasets, ensuring accuracy and reliability.
- Created and managed Regions of Interest (ROIs) for retail use cases and integrated configurations into the project database, showcasing problem-solving skills.

Karachi Development Authority, IT Department

IT Intern

Karachi, Pakistan

March 2025 – April 2025

- Supported IT infrastructure maintenance and database management to ensure system reliability and data accuracy, demonstrating a commitment to maintaining productivity.
- Collaborated with the IT team on process documentation and system improvements within government IT operations, indicating a willingness to work hard and learn.

PROJECTS

AI-Powered Financial Research Assistant | *None*

- Architected an agentic hybrid RAG system using FastAPI and LangGraph, combining unstructured semantic search (Pinecone) with structured XBRL financial fact retrieval (PostgreSQL) for accurate, reliable responses, demonstrating strong problem-solving skills and adaptability.
- Developed a semantic routing engine to classify user intent (quantitative vs. qualitative), dynamically grounding numerical queries in structured financial data to mitigate LLM hallucinations, showcasing commitment to producing high-quality results.
- Engineered a concurrent, just-in-time ETL pipeline using AsyncIO that detects missing SEC filings during live queries and triggers parallel scraping, parsing, and indexing (text and XBRL) for real-time responses, highlighting the ability to maintain productivity under pressure.

AI-Powered Factory Compliance System | *None*

- Developed a real-time computer vision system using YOLOv8 to detect PPE violations, machine occupancy, worker flow anomalies, and cleaning events, showcasing the ability to learn and apply new technologies quickly.
- Designed custom Regions of Interest (ROIs) with rule-based compliance tracking for precise, zone-specific monitoring, demonstrating keen attention to detail.
- Integrated modular detectors (helmet, glove, occupancy, cleaning) within a scalable architecture, logging violations, occupancy durations, and shift-wise summaries for automated compliance reporting, indicating a commitment to thoroughness and accuracy.

AI-Powered Face Recognition and Attendance System | *None*

- Developed a real-time facial recognition access system using MTCNN, FaceNet, and DeepSort with embedding-based matching via TensorFlow and DeepFace.
- Automated frame extraction, image augmentation, and database logging with OpenCV and OpenPyXL for efficient face-based access tracking.
- Designed a Streamlit dashboard for live monitoring and user registration, implementing modular OOP architecture with logging and exception handling.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, SQL

AI & Data Science: Computer Vision, Natural Language Processing (NLP), Retrieval-Augmented Generation (RAG), Data Science

Software Development: Object-Oriented Programming, Data Structures & Algorithms, Backend API Development

Other: Problem Solving